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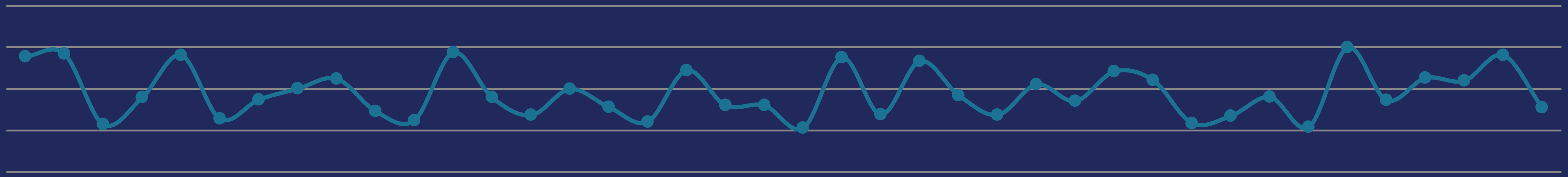
Time Series Analysis

Session 1 — Foundations



Prerequisites

What you should already be comfortable with before we start

Probability and Statistics
Distributions, expectation/variance, hypothesis testing

Linear Regression
OLS estimation, model assumptions, interpreting coefficients

Python Fundamentals
Comfortable writing and debugging basic scripts

pandas & notebooks
Loading, indexing, and plotting data in a Jupyter notebook

Course Content

A 6-week arc, from first principles to a final forecasting project

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Time Series Basics	**MA Models, Partial Autocorrelation, Notational Conventions**	**Using ARIMA models to forecast future values**	**Seasonal Models**	**Smoothing and Decomposition Methods**	**The Periodogram & project kickoff**

Grading

How your final grade is composed

10%

Participation

Attendance & engagement in session

40%

Assignments

Hands-on tasks & presentations

50%

Final Project

Applied forecasting project on real data

Communication

Where course information lives and how to reach us

Join our Team using this code:

Time Series Analysis Training Summer'26

uirnsga

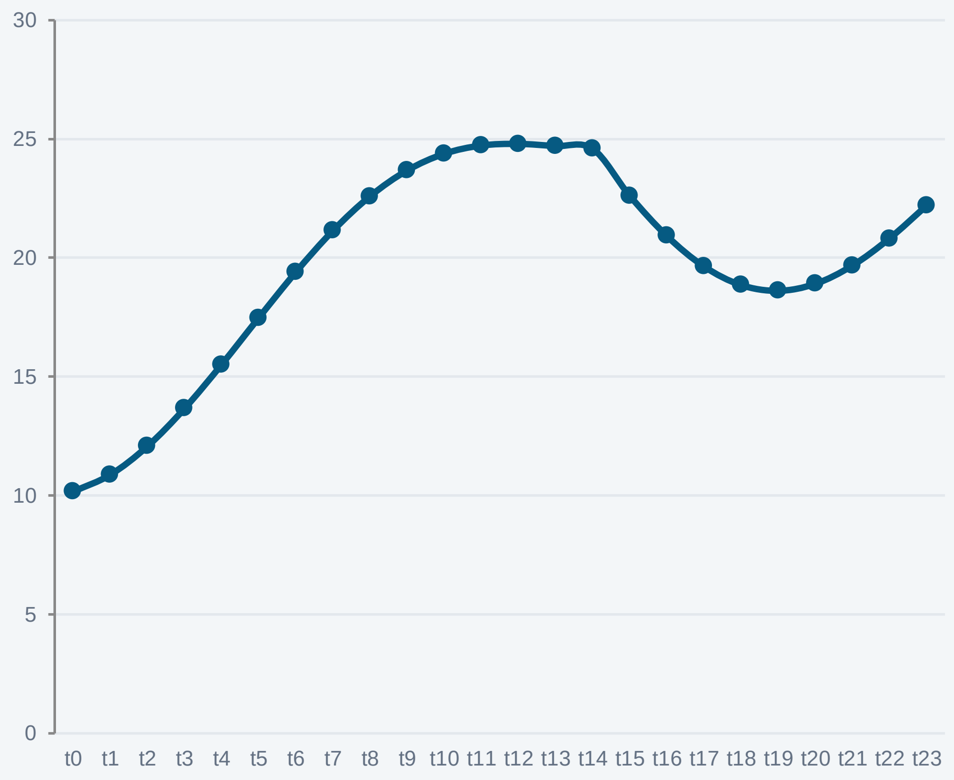
What Is a Time Series?

A univariate time series is a sequence of measurements of the same variable collected over time. Most often, the measurements are made at regular time intervals.

The key difference: order carries information. Unlike a typical regression dataset, you cannot shuffle the rows without losing something real.

- Daily stock closing prices
- Monthly unemployment rate
- Hourly temperature readings
- Quarterly GDP

A TIME SERIES, VISUALLY



Types of Models

Several families of models are used to describe and forecast time series

There are two basic types of "time domain" models.

- Models that relate the present value of a series to past values and past prediction errors - these are called ARIMA models (for Autoregressive Integrated Moving Average). We'll spend substantial time on these.
- Ordinary regression models that use time indices as x-variables. These can be helpful for an initial description of the data and form the basis of several simple forecasting methods.

Cross-Sectional vs. Time Series Data

Cross-sectional

- Many units, observed at one point in time
- Observations assumed independent
- Row order carries no information
- Standard OLS assumptions typically apply

Time series

- One (or few) units, observed over time
- Observations are typically autocorrelated
- Row order carries real information
- Needs methods that model dependence over time

Important Characteristics to Consider First

Some important questions to first consider when first looking at a time series are:

T

Trend

Is there a trend, meaning that, on average, the measurements tend to increase (or decrease) over time?

S

Seasonality

Is there seasonality, meaning that there is a regularly repeating pattern of highs and lows related to calendar time such as seasons, quarters, months, days of the week, and so on?

O

Outliers

Are there outliers? In regression, outliers are far away from your line. With time series data, your outliers are far away from your other data.

V

Constant Variance

Is there constant variance over time, or is the variance non-constant?

C

Long-run cycle

Is there a long-run cycle or period unrelated to seasonality factors?

A

Abrupt Changes

Are there any abrupt changes to either the level of the series or the variance?

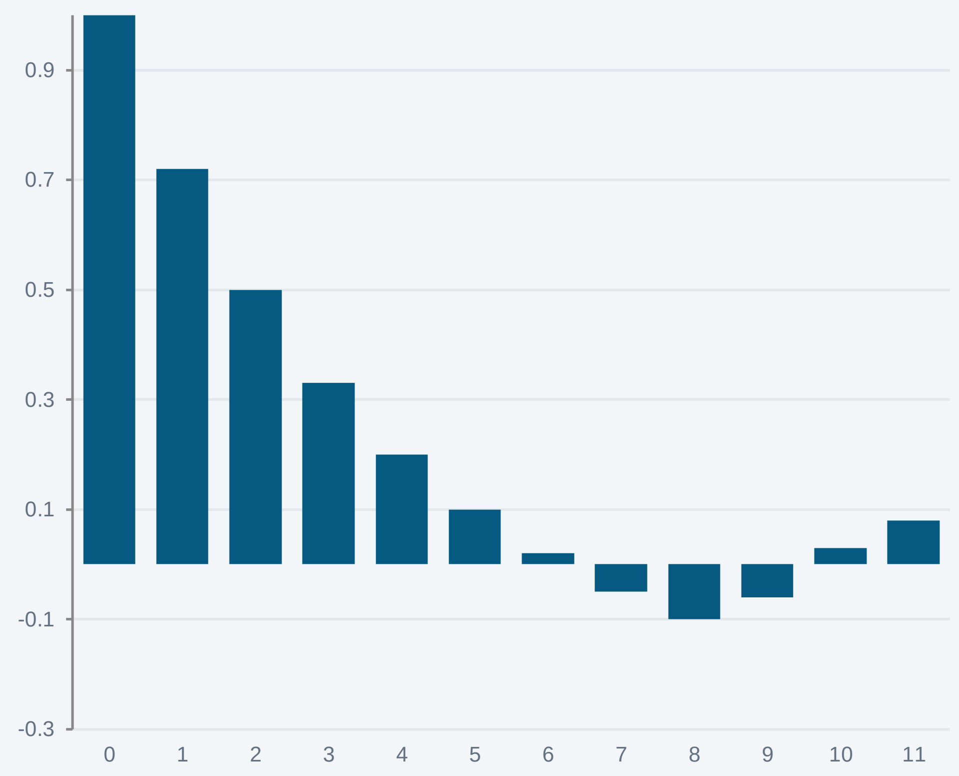
The Autocorrelation Function (ACF)

Measures how correlated a series is with its own past values, at each lag k.

$$\rho(k) = \frac{Cov(Y_t, Y_{t-k})}{Var(Y_t)}$$

Used to: spot repeating patterns, help choose model order, and check whether a series behaves like white noise.

A CORRELOGRAM (ACF BY LAG)



White Noise

A sequence of uncorrelated random variables and cannot be predicted.

μ

Zero mean

σ^2

Constant variance

Same spread at every point in time

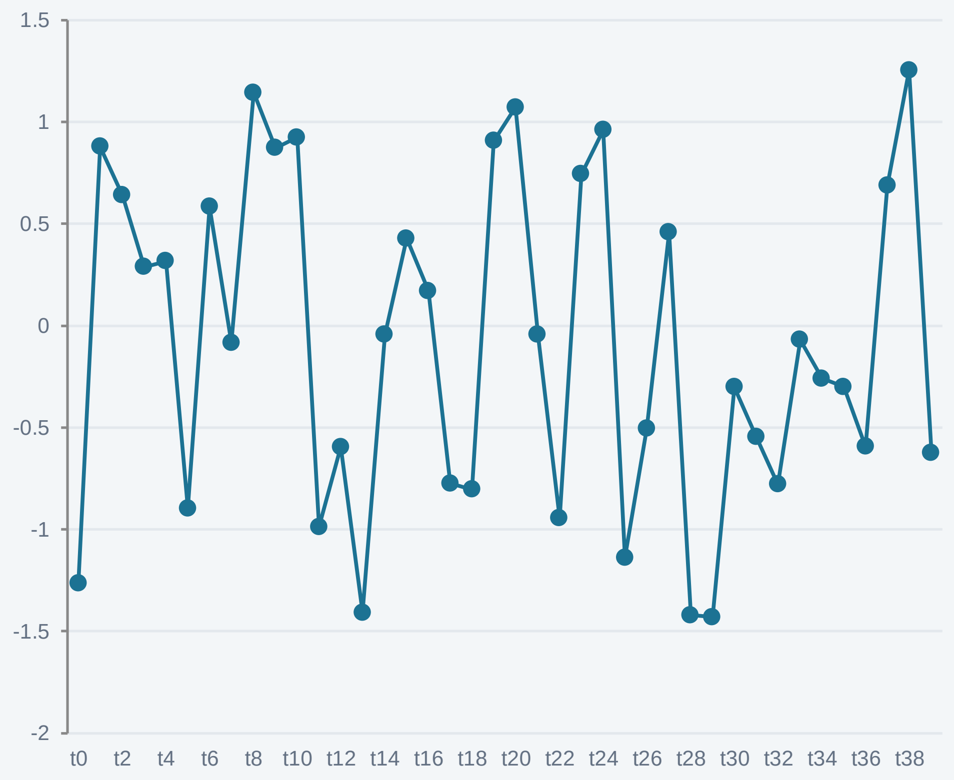
ρ

Zero autocorrelation

No relationship between values at any lag

Why it matters: *the residuals of a good model should look like white noise.*

WHAT WHITE NOISE LOOKS LIKE



Stationarity

Weak (covariance) stationarity requires three things to hold over time

μ

Constant mean

The average level doesn’t drift up or down over time

σ^2

Constant variance

The spread of values doesn’t grow or shrink over time

$\gamma(k)$

Lag-dependent covariance

Covariance between Y_t and Y_{t-k} depends only on k, not on t

Why it matters: most classical time series models assume stationarity. Non-stationary series are typically transformed before modeling.

Attendance

Summer'26: TSA Training Session 1